

Extending the Myanmar POLAR Dataset

Incorporating Task-3 and Multi-task Annotation

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Motivation & Background

- **Current Status:** The POLAR dataset includes Myanmar language data, but it is limited in size and lacks annotations for Task-3 (Manifestations).
- **Goal:** Extend the Myanmar POLAR dataset by collecting new sentences and annotating them for all three SemEval tasks.
- **Context:** As a co-author of both the ACL and SemEval papers, I aim to leverage our established frameworks to improve Myanmar NLP resources.
- **Action:** Engage 50 students to collect 600 sentences each (300 online, 300 offline) within 2 weeks.

SemEval-2026 Task 9

- Shared task on multilingual, multicultural, and multievent online polarization.
- Over 1,000 participants globally.
- 69 system description papers submitted.

The POLAR Dataset

- 110K+ instances across 22 languages.
- Addresses binary detection, type classification, and manifestation identification.
- Addresses gap in fine-grained polarization analysis.

Geographic & Linguistic Diversity

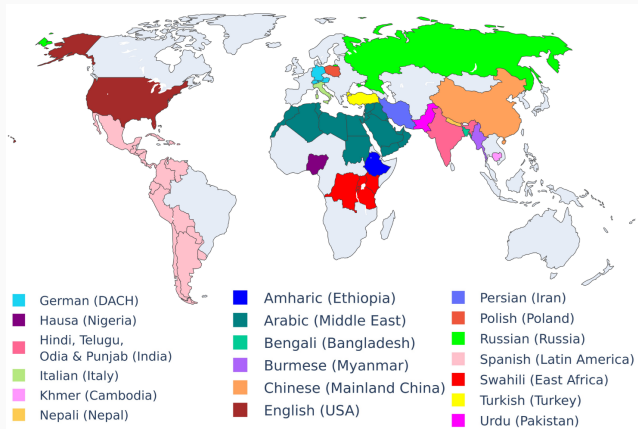


Figure 1: Languages represented in a world map covered by POLAR (Figure 1 from SemEval paper).

Paper Summaries

Summary of POLAR ACL Paper

Key Contributions:

- Released POLAR, a large-scale multilingual dataset.
- Defined a taxonomy of polarization types and rhetorical manifestations.
- Benchmarked Small Language Models (SLMs) and Large Language Models (LLMs).

Findings:

- Models perform well on binary polarization detection (Task 1).
- Performance drops significantly for Type (Task 2) and Manifestation (Task 3) prediction.
- Highlights the contextual and cultural complexity of polarization.

POLAR Construction Pipeline

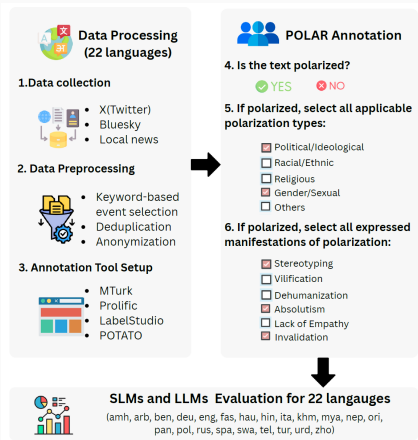


Figure 2: Pipeline for POLAR construction (Figure 1 from ACL paper).

Task Setup:

- Subtask 1: POLARDETECT (Binary)
- Subtask 2: POLARTYPE (Multi-label)
- Subtask 3: POLARMANIFEST (Multi-label)

Overall Top Performing Systems (Across All Languages):

- UTokyo Tsuruoka Lab, NYCU-NLP, SMASH.
- Common methods: Ensemble models, fine-tuning (LoRA), threshold tuning.
- Best models still struggle with low-resource languages like Khmer and Burmese.

SemEval-2026: Myanmar-Specific Leaderboard Results

SubTask 1: POLARDETECT

Rank	Username	Score
1	taien	0.8913
2	MKJ	0.8874
3	NYCU-NLP	0.8868
4	SMASH	0.8847
5	UTokyo Tsuruoka Lab	0.8836
6	mdok-style	0.8788
7	YEZE	0.8788
8	Lingo Research Group	0.8773
9	PSK	0.8771
10	MoMo	0.8763
–	<i>POLAR Baseline</i>	<i>0.8210</i>

SubTask 2: POLARTYPE

Rank	Username	Score
1	NASIM LAB	0.7474
2	SMASH	0.7358
3	UTokyo Tsuruoka Lab	0.7079
4	PolarMind	0.7015
5	DigiS-FBK	0.7000
6	PolaFusion	0.6988
7	YNU-HPCC	0.6988
8	NYCU-NLP	0.6938
9	YEZE	0.6920
10	UMUSP	0.6867
–	<i>POLAR Baseline</i>	<i>0.4772</i>

Current Status of Myanmar in POLAR

- **Data Size:** 4,334 instances.
- **Tasks Covered:** Only Task 1 (Detect) and Task 2 (Type).
- **Missing:** Task 3 (Manifestations) is entirely missing for Myanmar.
- **Inter-Annotator Agreement (IAA):** 0.13 (Fleiss' κ), indicating high subjective complexity.
- **Events:** Social issues regarding politics, ethnicity, and popular culture.

Our mission: Extend this dataset to include Task-3 and increase the data volume!

Task Definitions & Examples

Task 1: POLARDETECT

Definition: Determine whether a given text expresses attitude polarization (increasingly extreme, divided beliefs or behaviors between opposing groups).

English Example

Text: "Conservative values okay junior. Its an oxymoron, and hes just a moron."

Text: moron=ဦးနှောက်မရှိသူ၊ လူမိုက်၊ တုံးအသူ၊
oxymoron=အဓိပ္ပာယ်မတူသော စကားလုံးများကို တွဲသုံးခြင်း

Label: Polarization = 1

Myanmar Example

Text: ရှုံချ ပါ တယ် ပြည်သူ ကို လုပ် တာ နိုင် ရေး မ ဟုတ်
ပါ ဘူး အကြမ်းဖက် နှောင့်ယှက် တာ ပါ

(Trans: Condemn it. Doing this to the people is not for the cause, it's terrorist disruption.)

Label: Polarization = 1

Annotation Rule

Label as 1 (Yes) if the message clearly reflects attitude polarization. Otherwise, label 0 (No).

Task 2: POLARTYPE

Definition: If polarized, identify the social dimension underlying the polarization (Multi-label).

Categories:

- Political / Ideological (နိုင်ငံရေး / အတွေးအခေါ်ဝါဒဆိုင်ရာ)
- Racial / Ethnic (လူမျိုးရေး / တိုင်းရင်းသားရေး)
- Religious (ကိုးကွယ်ယုံကြည်မှုဆိုင်ရာ၊ ဘာသာရေး)
- Gender / Sexual (ကျားမ / လိင်အမျိုးအစား)
- Other (Economic, Media, etc.) (စီးပွားရေး၊ မီဒီယာ စသည်ဖြင့်)

Myanmar Example (Political=1, Other=1)

Text: စစ်တပ် က သတ် တာ ပြည်သူ တွေ မှ မ ဟုတ် တာ အကြမ်းဖက် တွေ လေ အဲ့ဒီ မသာ တွေ က

(Trans: The military shoots people, it's not the people, it's the terrorists, those dead ones.)

Task 3: POLARMANIFEST (Part 1)

Definition: Detect how polarization is rhetorically manifested.

1. Stereotype

(ပုံစံခွက်ထဲဝင် နေတဲ့ အတွေးအခေါ်)

Generalizing traits to all group members.

Mya: ကလေး ကို ကြည့် တာ နဲ့ အမေ ဖြစ် သူ ဘယ်လို စရိုက် လဲ သိ တယ်

(Look at the child and you know the mother's character.)

2. Vilification

(အပြစ်ဖို့၊ အသရေဖျက်၊ တိုက်ခိုက်ပြောဆို)

Defaming or demonizing a group.

Mya: အခြောက်မလင် ခွေးသူတောင်းစား
(Desperate beggar dog.)

3. Dehumanization

(လူသားဆန်မှု ကင်းမဲ့ခြင်း)

Stripping away human qualities (comparing to animals).

Mya: စစ်ခွေး သမီး ဆို တော့ ဖဲရိုက် မယ် xာခံ မယ် ဒါ ပဲ တတ် တာ မလား

(Military dog daughter, just shooting and sucking, is that all you know?)

4. Extreme Language

(အစွန်းရောက်တဲ့ အသုံးအနှုန်း)

Using absolute or dichotomous words ("always", "us vs them").

Mya: စောက်ရှူး တွေ အုပ်ချုပ် နေ သရွေ့ မ ကောင်း တဲ့ ဘက် သွား နေ မှာ ပါ

(As long as crazy cunts rule, it will only go to the

Task 3: POLARMANIFEST (Part 2)

5. Lack of Empathy (ကိုယ်ချင်းစာတရား ကင်းမဲ့ခြင်း)

Dismissing others' perspectives or experiences.

Mya: ငါ သာ ဆို အဆဲ တွေ ပဲ တန်း နေ အောင် ထွက် မှာ
(If it were me, I would just come out swearing.)

6. Invalidation (အသိအမှတ်မပြုခြင်း / တန်ဖိုးမထားဘဲ ငြင်းပယ်ခြင်း)

Denying a group's identity or existence.

Mya: မ နိုင် တာ သိ လို့ ဘုရင် စနစ် နဲ့ ကိုင် ပြီး ချုပ် နေ တာ
(Knowing they can't win, they suppress using the
monarchical system.)

Important Note for Students

Multiple labels are allowed! Texts often contain overlapping manifestations. Carefully read the context.

Data Extension Plan

Target: 600 sentences per student (Total: 30,000 sentences before filtering)

Online Sources (300 sentences)

- Facebook public pages
- News websites comments
- X (Twitter) / TikTok
- YouTube comments

Offline Sources (300 sentences)

- Books
- Magazines
- Novels
- Newspapers

Metadata Requirement

For EVERY sentence, you MUST record the original source information.

Why not CSV?

Manual CSV editing in text editors is prone to errors (comma escaping issues, misalignment).

Suggested Tools:

- Google Sheets / Excel (with data validation dropdowns for 0/1 labels).
- Label Studio (recommended for multi-label text classification).
- **POTATO** (Portable Text Annotation Tool, used in POLAR paper).

POTATO Quick Start:

We highly recommend using POTATO. It is open-source and easy to set up locally.

Docs:

<https://www.potatoannotator.com/docs/getting-started/quick-start>

GitHub: <https://github.com/davidjurgens/potato>

POTATO: Preparing the JSON Data File

POTATO requires input data in a JSON format. Each instance needs a unique `id` and the `text` to be annotated.

Example `data.json` (Showing 2 lines):

```
[
  {"id": "1", "text": "မြန်မာ့ သမိုင်း မှာ စောက်ရှက် မ ရှိ ဆုံး အယုတ်မာဆုံး ကို ပြ ပါ ဆို ရင် ဒီလို လူမျိုး တွေ  
ကို ပြ ရ မှာ ပဲ"},
  {"id": "2", "text": "စစ်တပ် က သတ် တာ ပြည်သူ တွေ မှ မ ဟုတ် တာ အကြမ်းဖက် တွေ လေ အဲ့ဒီ မသာ  
တွေ က"}
]
```

- Ensure the text is properly escaped if it contains quotes.
- The JSON file should be saved in your project directory (e.g., `data.json`).

POTATO: Configuring the YAML File (Part 1)

Basic File Setup (config.yaml):

```
annotation_task_name: "Myanmar POLAR Annotation"

# All relative paths below are resolved against task_dir.
task_dir: "."

# Data configuration
data_files:
  - "data.json"

# Tell Potato which fields in each record to use.
# id_key = unique identifier; text_key = the text shown to annotators.
item_properties:
  id_key: id
  text_key: text

# Where per-user annotation state is stored (relative to task_dir).
output_annotation_dir: "annotation_output/mya"
export_annotation_format: "json"
```

Key Configurations:

- Define your `annotation_task_name`.
- Set `task_dir` to your current working directory.
- Point `data_files` to your `data.json` file.
- Map `id_key` and `text_key` to match the fields in your JSON file.

POTATO: Configuring the YAML File (Part 2)

Annotation Schemes (config.yaml):

```
# Annotation scheme
annotation_schemes:
  # Task 1: Polarization Detection
  - annotation_type: radio
    name: polarization
    description: "Task 1: Is the text polarized? (1=Yes, 0=No)"
    labels:
      - "0"
      - "1"

  # Task 2: Polarization Type (Multi-label simulated via radio 0/1 for each)
  - annotation_type: radio
    name: political
    description: "Task 2: Political / Ideological (1=Yes, 0=No)"
    labels: ["0", "1"]
```

Key Configurations:

- Use `annotation_type: radio` for binary (0/1) choices.
- Create a separate scheme block for **each** of the 12 tags required (Task 1, 2, and 3).
- Set `require_password: false` at the very end of the file for easy student access.
- Notice how name matches the CSV column headers we defined earlier!

POTATO Installation: Environment Setup

Step 1: Create and activate a virtual environment

- Create environment: `python -m venv potato`
- Activate environment: `source potato/bin/activate`

Step 2: Install the POTATO package

- Install via pip: `pip install potato-annotation`

Step 3: Verify the installation

- Run command: `python -c "import potato; print(potato.__version__)"`
- Expected output: 2.7.1

Step 4: Set up the project directory

- Create folders: `mkdir -p polar/data && cd polar`
- Place your `data.json` inside the `data/` folder.
- Place your `config.yaml` inside the `polar/` folder.

Step 5: Directory structure should look like this:

```
polar/  
  config.yaml  
  data/  
    data.json
```

Step 6: Start the server

- Run command: `potato start config.yaml -p 8000`
- Open browser at: `http://localhost:8000`

POTATO: Annotation User Interface

How to Annotate:

- The text to be annotated is displayed at the top.
- Annotators select the "0" or "1" radio button for each tag across Task 1, Task 2, and Task 3.
- Use the navigation buttons at the bottom to move to the next instance.
- Login requires only a username (e.g., alice).

The screenshot shows a web browser window with the title 'Myanmar POLAR Annotation'. The address bar shows 'localhost:8020/annotate'. The page has a purple header bar with the title 'Myanmar POLAR Annotation' and a status bar showing 'Not labeled', 'Progress: 0/10', 'Instance: #1', and navigation buttons. The main content area is divided into two sections. The top section, 'Text to Annotate:', displays a Burmese text snippet: 'မြန်မာ့ သမိုင်း မှာ စောက်စက် မရှိ ဆုံး အယုတ်မာဆုံး ကို ပြု ပါ ဆိုရင် ဒီလို လူမျိုး တွေ ကို ပြု ရ မှာ ပဲ'. The bottom section contains eight tasks, each with two radio buttons (0 and 1). The tasks are: Task 1: Is the text polarized? (I=Yes, O=No); Task 2: Political / Ideological (I=Yes, O=No); Task 2: Racial / Ethnic (I=Yes, O=No); Task 2: Religious (I=Yes, O=No); Task 2: Gender / Sexual (I=Yes, O=No); Task 2: Other (I=Yes, O=No); Task 3: Stereotype (I=Yes, O=No); Task 3: Vilification (I=Yes, O=No); Task 3: Dehumanization (I=Yes, O=No); Task 3: Extreme Language / Absolutism (I=Yes, O=No).

Figure 3: Screenshot of the POTATO annotation interface.

Required CSV Format (Final Submission)

If you use Google Sheets or convert POTATO's JSON output, ensure your final data matches this schema:

Field	Description
id	Unique identifier
text	The collected sentence
source_type	online / book / etc.
source_detail	URL or Book/Page
polarization	1 or 0 (Task 1)
political	1 or 0 (Task 2)
racial	1 or 0 (Task 2)
religious	1 or 0 (Task 2)

Field	Description
gender	1 or 0 (Task 2)
other	1 or 0 (Task 2)
stereotype	1 or 0 (Task 3)
vilification	1 or 0 (Task 3)
dehumanization	1 or 0 (Task 3)
extreme_lang	1 or 0 (Task 3)
lack_of_emp	1 or 0 (Task 3)
invalidation	1 or 0 (Task 3)

Timeline & Conclusion

Action Items for Students:

- Read the ACL and SemEval papers carefully.
- Learn the existing Myanmar POLAR dataset format.
- Try to understand the definition of each tag thoroughly.
- If you want to use the POTATO annotation tool, set it up based on the GitHub link and quick start guide. (JSON and YAML templates will be provided).
- Collect and label sentences carefully, ensuring all source metadata is tracked.

Deadline

23 August 2026 (Sunday)

All 600 sentences and annotations must be submitted by this date.

Conclusion & Future Work

- **Immediate Outcome:** A newly extended Myanmar POLAR dataset with Task-3 annotations.
- **Research Impact:** First step towards comprehensive fine-grained polarization analysis for the Myanmar language.
- **Future Experiments:**
 - Run baseline experiments using mBERT, XLM-R, and LaBSE on the extended data.
 - Compare SLM vs LLM performance specifically for Myanmar.
 - Analyze cultural specificities in Myanmar polarization manifestations.

Questions & Discussion

References

- Naseem, U., et al. (2026). POLAR: A Benchmark for Multilingual, Multicultural, and Multi-Event Online Polarization. In *Proceedings of ACL 2026*.
- Naseem, U., et al. (2026). SemEval-2026 Task 9: Detecting Multilingual, Multicultural and Multievent Online Polarization. In *Proceedings of the 20th International Workshop on Semantic Evaluation (SemEval-2026)*.
- POLAR Trial Data Repository: <https://github.com/Polar-SemEval/trial-data>
- POLAR Leaderboards: <https://github.com/Polar-SemEval/Leaderboards>
- SemEval-2026 Task 9 Starter Notebook:
<https://github.com/Polar-SemEval/SemEval2026-task9/blob/main/starter.ipynb>
- POTATO Annotation Tool: <https://github.com/davidjurgens/potato>